Fairness in Network Anonymization

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ABSTRACT

We define a kind of fairness for network anonymization, and propose some methods to measure it. We investigate the fairness with regard to different sized communities of 19 network anonymization methods. We do experiments on real world and synthetic networks, and find that unfairness is heavily dependent on the anonymization method, with different algorithms favoring community sizes.

Keywords: Fairness, Network Anonymization, k-Anonymity, Differential Privacy

1 INTRODUCTION

Network Anonymization is the process of attempting to hide sensitive information about the nodes or connections in a network from malicious actors. Sensitive information may be the identity of a node, the participation of certain users, or the existence of certain links between users. In many social networks, users publish information about themselves, which may then be revealed to malicious actors if the network is released for analysis. In general when networks are published for analysis, the identities of the nodes are removed. However, this does not guarantee privacy, as many de-anonymization procedures can use the structure of a network to re-identify nodes Ji et al. (2017). To provide privacy for the individuals who make up a network, this potential for structural identification must be taken into account when networks are published. There are a number of ways to address this issue, many of which are detailed in Majeed and Lee (2021). One common privacy technique is k-anonymity, which limits adversary ability to re-identify a node based on its structural attributes. K-anonymity comes from database anonymisation, and guarantees that for each entry, there are at least k entries that are indistinguishable de Jong et al. (2025), so that there is only a $1/k$ chance of an adversary identifying an entry based on the attributes. In graphs, k-anonymity is based on some node metric, such as the degree, or neighbourhood structure. Another common anonymity target is differential privacy, which aims to hide the participation of particular nodes or edges in a network. This also originates from database anonymisation, and is used in situations where queries are used to obtain statistics for data analysis. A database satisfies the differential privacy condition if it is not possible to determine the existence of any particular entry from query results Sala et al. (2011). This is usually done by adding random noise to query results. In graph data, differential privacy is achieved when the participation of a node cannot be determined from the anonymous graph (Sala et al. (2011)). There are also a number of other less popular privacy targets that algorithms may aim to achieve (Wagner and Eckhoff (2018)).

In Social Network Analysis (SNA), fairness is the idea that all types of nodes should be treated equally. Social networks mirror real world interactions and patterns of behaviour, so real world communities and groups are mirrored in online social networks. Because of the fact that people interact more with people like them (called homophily in networks), these communities and groups are often split along categories such as gender, race, or language. It is important for social network analysis and interventions to produce equal quality results for all users, thus the investigation into fairness. Fairness has been studied in many areas of SNA. Many traditional network analysis techniques do not actively target fair results, and have been found to produce results that unfairly disadvantage certain communities (Saxena et al. (2024)). In network anonymisation fairness has not yet been investigated, and since anonymisation is always involved in responsible network data publishing, it is critical for anonymisation procedures not to introduce new biases, or strengthen existing ones.

1.1 Contributions

In this paper we implement, and compare the fairness of 19 different network anonymisation methods. We focus on fairness with regards to the size of labeled communities. We evaluate the anonymisation algorithms by measuring the anonymity, and utility of the anonymous graphs they produce.

1.2 Thesis Overview

In section 2 we present some related works on the subject of network fairness and network anonymization. In section 3 we present how we will evaluate the anonymisation algorithms. section 3.2 presents the graphs that we chose to use for testing, and provides information about their structure. section 3.3 we detail how we implemented the anonymisation algorithms, and the evaluation measures. In **??** we present the anonymization methods that we implement. In **??** we present the anonymity and utility scores of the different anonymization methods, and evaluate their fairness.

2 RELATED WORKS

In this chapter we outline related work on graph anonymisation, and fairness in SNA. We focus on how graph anonymisation algorithms are distinguished, and how to measure their effectiveness. For fairness we consider what types of fairness have been proposed and how they are measured.

Because communities have structural differences, SNA techniques perform differently on them, which can cause minorities to be unfairly disadvantaged. Saxena et al. (2024) is a comprehensive survey of the different types of fairness that have been investigated in SNA. The work distinguishes between individual and group as types of fairness, where individual fairness considers differences in the treatment of similar users. Group fairness considers inequality between different communities that exist in a network. The targets for fairness they outline are equity, equality, and statistical parity. Equity for groups is defined as different groups performing the same relative to their size. Equality for groups is defined as different groups receiving the same resources relative to group size. Parity for groups is based on the variance of results over different groups. Saxena et al. (2024) details fairness research in link prediction, centrality measurement, influence maximization and minimization, and community detection. They also highlight areas of SNA where fairness has yet to be investigated, among which is network anonymisation.

There are a number of different ways that anonymisation algorithms have been evaluated and categorized. Majeed and Lee (2021) classifies anonymization algorithms by what information they protect and by the anonymizing mechanisms. They assess a set of 46 graph anonymization algorithms, and note their anonymization mechanism as well as their merits and demerits with regard to anonymity, utility, complexity, and scalability. They give a detailed classification of different types of anonymity, graph utility, privacy threats, and anonymization operations. Zhou et al. (2008) provides ways to classify anonymization algorithms by the anonymizing mechanisms. For some algorithms they detail what information is protected by the anonymization, and what utility is preserved.

The evaluation of anonymisation algorithms can be done by way of some anonymisation metric. Ji et al. (2017) investigates several anonymization algorithms with regard to their resilience against different types of de-anonymization attacks. They also examine the information utility that is preserved by anonymization methods, and detail many ways to measure the graph utility. Wagner and Eckhoff (2018) is a compilation of different metrics to assess anonymity for different types of data, including social network data. They characterize the metrics based on what aspects of the graph or adversary they measure. They also detail for each metric what aspects of privacy they consider, and the information required to compute them.

Sala et al. (2011) defines differential privacy for graph data using the DK series of a graph as the query function. They propose creating a differentially private graph by adding LaPlace noise to the 2K series or joint degree series, and publishing a graph constructed from the noised series. They provide a proof of ε-differential privacy for their anonymization mechanism. de Jong et al. (2025) defines several k-anonymity measures on graph data, and compares their anonymization strength and their computational efficiency.

The combination of fairness and network anonymisation remains unexplored. There are some algorithms that aim for a certain type of fairness. For instance Zheng et al. (2020) proposes a method for network anonymisation that is fair to users' particular privacy settings, so that users who share more personal information recieve better services than users who share less personal information. However there are no algorithms that focus on fairness across different user groupings.

The goal of this work is to start investigating fairness with regard to different communities in social network anonymisation.

3 METHODOLOGY

3.1 Preliminaries

We define some graph terms and properties that we will use in our experiments. A network $G(V,E)$ is a set V of n nodes, and a set E of m edges between those nodes. In this paper we refer interchangeably to networks and graphs. We use both vertices and nodes to refer to nodes, as well as edges or links to refer to edges. A directed graph is one where the edges have a direction. In an undirected graph edges do not have a direction, and (u,v) is equivalent to (v,u). In a weighted graph the edges have a weight value associated to them, to signify for instance length, or frequency of an edge. Simple graphs are those that are unweighted and undirected, where each edge occurs only once, and nodes do not connect to themselves. We focus on anonymisation algorithms designed for simple graphs. Communities in a social network are groups of nodes created by some method. In this work we use communities that are created by grouping nodes based on their attribute values. Node attributes are values associated to a node, in social networks these might be a user's age, location, or gender.

The degree sequence of a graph is the list of the degrees of its nodes sorted from highest to lowest. The joint degree sequence of a network is the sequence of degree pairs for each edge. This is also sorted from highest to lowest, first by maximal degree in the pair and then by minimal degree in a pair Sala et al. (2011). The d-neighborhood of a node is all nodes that can be reached by at most d edges from the node. A random walk starting from a node u selects a random edge (u,v) connected to u and moves to its endpoint v. This is repeated for the length of the walk. The Jaccard Similarity of two nodes is computed as the Jaccard Similarity of their 1-neighbothoods. So $JS(u,v) = \frac{N_1(u)hatN_1(v)}{N_1(u)UN_1(v)}$ that is the number of neighbors they have in common, divided by their combined number of neighbors. [cite JS?] The betweenness centrality (BC) of a node is the fraction of shortest paths that pass through it. We compute the betweenness centrality using networkx *nx.betweenness_centrality()* Hagberg et al. (2008). We use the distribution of the betweenness centrality, and the nodes with the highest betweenness centrality as measures of the structure of a network. The local clustering coefficient (LCC) of a node is the fraction of its neighbors that are connected to each other. We compute this using networkx *nx.clustreing()* Hagberg et al. (2008). Like betweenness centrality we use the distribution of the local clustering coefficient as a measure of the structure of a network. The number of changed edges (CE) of a node is the sum of the number of edges removed and added in the anonymisation process. We compute $CE(u)$ as the number of neighbors that are different between the neighbors of u in the ground truth graph and the neighbors of u in the anonymous graph.

The k-anonymity measures for a graph the minimal value of k, that is the size of the equivalence class for a certain node property. For each node the value of the node property is calculated, and nodes are in the same equivalence class if they have the same value of the given property. Degree is a common node property used for k-anonymity. For a node with k-degree anonymity, an adversary has a $1/k$ chance of identifying the node based on its degree.

Differential privacy hides the participation of a node or edge in a network. "An ε-differentially private graph system would output a graph that given a statistical description of an input graph, the probability of seeing two similar graphs as the real input graph is close, where closeness between the two probabilities is quantified by ε. A larger value of ε means it is easier to identify the source of the graph structure, which means a lower level of graph privacy" Sala et al. (2011). This is designed for interactive database privacy, where the existence of a user should be impossible to determine from query results Sala et al. (2011).

The Kolmogorov-Smirnov (KS) test is a statistical test used to evaluate the equivalence of two distributions. We compute this using the scipy's kstest *scipy.stats.kstest()* Virtanen et al. (2020). The KS test returns the maximum vertical distance between two cumulative distributions. We use this to measure the difference between node property distributions in the ground truth graph and the anonymous graph.

3.2 Graph Details

We ran our tests on both synthetic and real world networks. The minority and majority split was created by partitioning the nodes of the graphs based on some node attributes. For the synthetic graphs the attributes were labels for communities with predetermined properties. For the real world networks we used existing

node information such as gender or ethnicity, and give a description of the structural attributes. For the rest of this paper we refer to the partition groups as communities, where those with more than half the nodes are the majority, and those with fewer than half the nodes are the minority.

Using synthetic networks allows us to have control over the community properties. We can directly determine the size of the communities for a clear majority and minority. We chose to construct our synthetic graphs using the Hierarchical Barabasi Albert (HBA) method Saxena et al. (2023). HBA graphs are generated by iteratively adding nodes and edges. Nodes are assigned a community, and edges are created within or between communities at random based on a specified homophile factor h. We choose to use graphs with 10 000 nodes, and two communities, with 80 and 20 % of the nodes respectively. The homophily factor that we used was 0.2. [get the full set of variables that we used for the graph] We also used the set of obesity networks from Wilder et al. (2018). This is a set of synthetic networks created using a simulation of a real world community. The graphs each have 500 nodes, and geographic location, age, ethnicity, gender, and weight status as node properties.

One real world network is a collection of 4039 Facebook users where the ”friendships“ form the connections McAuley and Leskovec (2012). The minority / majority community split here is formed by the gender attributes of the users. About 38% of nodes have gender attribute 1 and about 62% have gender attribute 0. This gives a distinct minority and majority group, although the difference in size is not as pronounced as in the synthetic dataset.

Another real world network is a collection of twitter interactions Macedo and Saxena (2024). The connections between users are based on ”retweets“. The network has 367188 nodes and 816869 edges. Since ”retweets“ are directional, this is a directed network. Since we focus on undirected networks, we regard each connection as bidirectional and flatten the network to an undirected graph. The network had too many nodes for us to anonymise the full graph, so we used random sampling to get smaller graphs. We sampled the graph by adding each edge from the full graph with a probability of 0.015. Sampling based on the edges allowed us to get graphs without nodes of degree 0.

We computed the degree distributions of the labeled communities within each graph. We measured the homophily factor by computing the fraction of within-group edges for each community. We also measured the ability of modularity based community detection to reproduce our labeled communities. For each labeled community we compute the likelihood that any two of its nodes are assigned the same group by the modularity based community detection.

We found that the soccer graphs have a community size split very similar to 80/20, and that the Facebook gender split is very close to 60/40. We found that when comparing communities of similar sizes, we found the Facebook Gender split, HBA graphs, and the Soccer network had a similar homophily factor. The obesity graph, and Facebook Multicommunity communities had a higher homophily factor relative to the community size.

The degree distribution is different for the HBA graphs than for the other networks. The HBA graph levels off at a higher fraction of the maximum degree than the fb Gender network. There are more nodes with intermediary degrees in the HBA network than in the fb Network. The degree distribution is consistent with the real world networks in the obesity graphs.

3.3 Experimental Setup

3.3.1 Implementation of the Algorithms

In implementing the algorithms, we stayed as close as possible to the papers' descriptions of the methods. All of the anonymisation algorithms output a single anonymous graph. All of the algorithms were implemented using python 3.11.0. The exception to this is ANO-NET de Jong et al. (2024), where we used the original C++ code, and made modifications to save the graphs to file. We use the Networkx python package to create and store the graphs Hagberg et al. (2008).

We implemented the metrics for measuring utility, and the differential privacy metric using Python 3.11.0, and Networkx. We computed the k-anonymity partitions using the C++ code from ANO-NET de Jong et al. (2024).

3.3.2 Details on Running Testing

Many of the anonymisation algorithms have a probabilistic element, so we do a series of five runs for each algorithm. The experiments were run on an Apple MacBook Pro with an Apple M2 Pro chip, running macOS Sonoma 14.4.

3.3.3 Fairness Assessment

We assess the fairness between different community sizes. We consider the communities based on the labels of the nodes. We consider an algorithm fair in a certain metric if all communities in the anonymous graph it produces have the same score in that metric. The communities in the anonymous graph contain the same nodes as the communities in the ground truth graph.

3.3.4 Metrics

We want to consider both the anonymity and the utility of the anonymous graphs. For the anonymity we chose to use k-anonymity, and differential privacy to assess the privacy of each node. To assess the utility of the anonymous graphs, we compare the structural properties of the ground truth graph to those of the anonymous graphs.

We measured 5 different kinds of equivalence. Degree considers the degree of each node. Count considers the number of nodes, and the number of edges in the neighborhood of each node. Vertex refinement query (VRQ) considers the degrees of the nodes in the neighborhood of each node. Isomorphism considers the full neighborhood structure of each node de Jong et al. (2025). We used the code by de Jong et al. (2024), to compute the partitions based on these equivalence types, and modified it to output them for later analysis.

To measure differential privacy there was no pre-defined metric. We base our analysis on the definition of differential privacy for graph data from Sala et al. (2011). They define edge differential privacy on graphs by stating that an anonymisation algorithm satisfies ε differential privacy if the following holds,

$$|\ln \frac{P[\tilde{DK}(G) = s)]}{P[\tilde{DK}(G^*) = s)]}| \leq \varepsilon \tag{1}$$

where $\tilde{DK}$ is the privacy mechanism performed on the dK series, s is a possible output of $\tilde{DK}$, and G and G^* differ by at most one edge. In order to make this easier to compute we estimate the likelihood of two graphs being related in this way by the similarity of the joint degree distributions. We use this estimation to compute what we call a "Differential Privacy Score". The formula for this is given below.

$$DP(v_0) = \max\left(\frac{d(D2(G), D2(G'))}{d(D2(G \setminus v_0), D2(G'))}, \frac{d(D2(G \setminus v_0), D2(G'))}{d(D2(G), D2(G'))}\right) \tag{2}$$

where G is the ground truth graph, G' is the anonymous graph, and $D2(G)$ is the joint degree series of G.

To assess the utility of the anonymous graphs we consider the similarity of the anonymous graphs to the ground truth graph in several graph properties. The betweenness centrality gives information about the relative importance of nodes in the graph structure. [we should consider how many of the top 10 / 20 are in the min vs maj] We compute the betweenness centrality using *nx.betweenness_centrality()* and use $\log(n) + 1$ pivots. Using the pivots makes the computation much faster, and the results are still very consistent. The local clustering coefficient (lcc) is computed using *nx.clustering()*. We compute the number of changed edges per node by adding the number of new edges, and the number of removed edges.

3.3.5 Performance Analysis

3.4 Anonymisation Algorithms

Most of the anonymisation algorithms that we investigated can be broadly separated into k-anonymity based algorithms, and differential privacy based algorithms. There are a number of algorithms that are neither (Wu et al. (2019),Yan et al. (2018),Beato et al. (2013),Mittal et al. (2012),Liu and Mittal (2016),Kumar and Kumar (2017)), and most of these are focused in some way on preserving link privacy, rather than node privacy. Since all of the k-anonymity algorithms that we chose use the minimal equivalence class size (k) as a parameter, we used 10 in all algorithms for the sake of consistency. The differential privacy and link privacy anonymisation algorithms have much more varied parameters, so we used the parameter values suggested as the best by the papers that introduced the algorithm.

3.4.1 ANO-NET

The k-anonymization procedure proposed by de Jong et al. (2024) can use several different forms of anonymity. The overall structure of the algorithm is as follows. For each of the edges a specified measure

Anonymisation Algorithm	**Degree Sequence**	**Reconstruct by Creating Edges**	**Add edges**	**Remove edges**	**Add nodes**
Naive					
ANNM	Partition		Deterministic		Yes
ANONET AffU edge selection				Weighted Random	
DPFT	Add Noise	Weighted Random			
edgeEnt		Deterministic			
ENDP		Deterministic			
FiM			Random	Random	
kDegSeq	Partition		Random	Random	
KDVEM	Partition		Deterministic		Yes
linkMirage			Weighted Random	Random	
linkPerturb			Weighted Random	Random	
OLP				Deterministic	
PBCN	Partition + Add Noise	Preferential			Yes / Sometimes
pseudoNode	Partition		Deterministic		Yes
randomWalk			Weighted Random	Weighted Random	
roughSet		Deterministic			
subgraphDP		Weighted Random			
twoStep	Partition + Add Noise	Deterministic			

Table 1. An overview of the mechanisms used in the considered anonymisation algorithms.

is computed to determine how much the edge affects anonymity. A specified number of the edges that have the highest impact are removed. This is repeated until a certain budget of removed edges is reached.

We choose to use 10% of the edges as the total budget. The edge selection method that we choose to use is referred to in the code as AFF-U. In this method, the number of non-anonymous vertices that are affected by removing an edge are computed. The edges to be removed are selected probabilistically, with the edges that impact the most nodes most likely to be removed. We chose set the parameter to recompute this after each 10% of the budget is used. We use this same setup for each of the k-anonymity types we test. We chose to anonymise graphs using degree, count, and VRQ as the equivalence set. For each of these we used only the 1-neighborhood of each node. For this algorithm we use the C++ code by de Jong et al. (2024). We modified the code such that it outputs the anonymous graphs.

3.4.2 A New Method for Adding Noise Nodes

A New Method for Adding Noise Nodes, (ANNM) is proposed by Hamzehzadeh and Mazinani (2019) as an anonymization algorithm to achieve k-degree anonymity. The algorithm uses the degree sequence of the graph, and groups by degree such that each group has the same degree value. Then classify groups as secure and insecure, and insecure groups are assigned priority 0 or 1. Target groups are created one at a time by adding nodes from priority 0 insecure groups to priority 1 insecure groups. To achieve the target degrees, edges are added between nodes in the same group based on betweenness centrality. If the existing nodes are not sufficient for this, dummy nodes are added.

3.4.3 k-Degree Sequence Anonymisation

The anonymisation algorithm is proposed by Casas-Roma et al. (2016) for k-degree anonymity. It works by creating a k-anonymous degree sequence, and modifying the edges to fit. The degree sequence is grouped by univariate microaggregation (UMGA) into groups of size at least k. For each group the target degree is either the floor or the ceiling of the average degree. The optimal solution is found using greedy optimization for the smallest absolute value of the sum of degree differences over all groups. The best solution is saved, and when no improvement is made over 20 iterations, the solution is returned.

Based on the target degrees, the degree deficit or surplus is calculated for each node. If overall edges need to be added, two unconnected nodes with a deficit are selected at random, and are connected. If overall edges need to be removed, two nodes with a surplus, and one neighbor of each of them are selected at random. The edges between the surplus nodes and their neighbors are removed, and the neighbors are connected to each other. If no edges need to be added, a node with a surplus and one with a deficit are chosen. One of the neighbors of the surplus node is disconnected, and connected instead to the deficit node.

3.4.4 Pseudo Node Anonymisation

Pseudo Node or Dummy Node anonymisation is proposed in Chester et al. (2012) for k-degree anonymity. The edges of the original graph are not changed, rather the graph is anonymised by adding connections to new nodes. The degree sequence of the graph is partitioned into sets of vertices of at least size k, and the maximal degree in each group is taken as the target degree. A number of dummy nodes are added to the graph, and are connected to the old nodes until all degree deficits are satisfied. The new edges are added by iterating through the new vertices in order, and through the original vertices in order of largest degree. This continues until all of the original vertices have their target degree. Because of the way the new edges are added, the dummy nodes all have one of two degree values, and are k-anonymized by connecting to each other.

The partitioning of the degree sequence is done by using a recursive algorithm that returns the cost and the optimal starting element for a partition group when given the last element. The optimization is done with regard to the maximum deficiency in node degree between the original and k-anonymous degree sequence, since the maximum deficiency gives a tight lower bound on the number of new vertices that need to be added.

3.4.5 k-Degree Anonymity with Vertex and Edge Modification

k-Degree Anonymity with Vertex and Edge Modification (KDVEM) is proposed by Ma et al. (2015) for k-degree anonymity. It partitions the degree sequence into groups with size at least k, using greedy minimization of the total degree deficit using the maximal degree in each group as the target degree. For each vertex, candidates for connection are selected based on distance and greedy modularity community. Edges are added in order between nodes and candidates until all degree deficiencies are satisfied. If there

are then still nodes that are not sufficiently anonymous, dummy nodes are added and anonymised as in Pseudo Node Anonymisation.

3.4.6 Two-Step Anonymisation

Two-Step anonymisation is proposed by Liu and Terzi (2008) for k-degree anonymity. This algorithm partitions the degree sequence in the same manner as in Pseudo Node anonymisation, but with regard to the total degree deficiency, rather than maximal. The algorithm then tries to reconstruct graph based on the sequence, randomly increasing degrees if the sequence is not realisable. The graph is realized by choosing a random node, and connecting it to its neighbours highest degree first, and then to other nodes highest degree first. If the target sequence is not realizable, then uniform random noise between 1 and 5 is added to the smallest degree, and the partition step is redone.

3.4.7 Differential Privacy with Field Theory

Differential Privacy with Field Theory (DPFT) is proposed by Zhu et al. (2019) to achieve ε-differential privacy. It computes the correlation between each pair of nodes as the sum of the transition probabilities for each length of random walk up to 5. The distance between two nodes is taken to be the inverse of their correlation. For each node a list of candidate nodes that have a non-zero correlation is compiled. For each node candidate pair, the product of their degrees, and some exponentiation of their correlation, called force, is saved. In the computation of the force we chose to use exponent 1 for speed and ease. A target degree sequence is then constructed by adding Laplace random noise with parameter 2 to each degree in the original sequence.

To create the anonymous graph, the algorithm builds a queue of nodes, and a list of edges to add to the graph. A node is randomly selected with probability proportional to its perturbed degree. If its candidates list is non-empty, a candidate node is chosen with probability proportional to the force between the two nodes. Each node is removed from the other's candidate set. If the edge between them is not already in the edge list, it is added, and the degree of each node in the target degree sequence is decreased by one. If the node's candidates list is empty, the vertex is added to the queue. This is repeated until all vertices have been added to the queue. There may be vertices left that do not have a sufficient number of edges in the edge set. Additional edges are added to these by checking for each node in the queue whether there are edges that can connect to one of the insufficiently connected nodes. If there are, they are added to the edge list. The algorithm outputs a graph with the nodes from the input graph, and the edges from the generated edge list.

3.4.8 Edge Insertion Differential Privacy

Edge Insertion Differential Privacy (ENDP) is proposed by Macwan and Patel (2018) for ε-differential privacy. The node list is sorted from lowest to highest degree. The anonymous graph is created by taking the same nodes as in the ground truth graph, initially all with degree 0. From the lowest degree node edges are added connecting to its original neighbors, also beginning from the lowest degree nodes. Edges are only added if the current degree of both nodes is below a threshold value θ. In our implementation we used as a threshold value the 80th quantile of the original degrees.

3.4.9 Privacy Preservation Based on Clustering and Noise (PBCN)

Privacy Preservation Based on Clustering and Noise (PBCN) is proposed by Huang et al. (2020) for ε-differential privacy. The algorithm does k-means clustering of the degree sequence into 10 groups, and finds the average degree of each cluster. The groups are perturbed by randomly selecting edges, selecting two clusters by applying Laplace noise with parameter $\frac{\lceil m/100 \rceil}{0.033}$ where m is the number of edges, to the cluster assignments of the original endpoints, and connecting randomly selected nodes from each selected cluster. Laplace noise with parameter $\frac{S(f)_i}{0{,}033}$ where $S(f)_i$ is the sensitivity of group i, is added to the degree of each node, using the sensitivity of the clusters. The graph is then reconstructed using *nx.havel_hakimi_graph()*. As post processing, based on Laplace noise with parameter $\frac{1}{0{,}033}$, either a number of noise nodes are added, and connected to true nodes, or a number of nodes within the bottom 5% of degrees are removed.

3.4.10 Subgraph DP

Subgraph Differential Privacy is proposed by Nguyen et al. (2015) for ε-differential privacy. A set of real and false edges are created, and each is assigned a probability, and the graph is constructed by sampling edges with these probabilities. Edges are selected using a breadth first search (BFS) of the 5-neighborhood

of each node. We select the real edges, and half again as many false edges, with nodes chosen from the BFS tree. For each of the selected edges, their importance is computed using the "normalized mutual friends" metric which is computed by the common neighbors of the nodes, divided by the sum of their degrees, and then rescaled to better preserve the average node degree. Laplace noise with scale $\frac{-1}{\ln(\frac{2}{e^{1/2}+1})}$ and location 0 is added to the weights, and the graph is reconstructed by taking those edges with weight smaller than 0.

3.4.11 Optimisation via Link Perturbation using Link Importance

Optimisation via Link Perturbation using Link Importance (OLP) is proposed in Wu et al. (2019). The target of the algorithm is to obscure the link relationships between nodes, which they measure via accuracy of structure prediction. The algorithm achieves this by removing the most regular edges in the graph. They compute the most regular edges by choosing 20% of the nodes at random, and from each of these nodes doing a random walk of length 100. The links in the walk are tracked, and each time an edge is used, its importance increases. The top 10% of edges by this metric are assumed to be the most regular, and are removed.

3.4.12 Edge Entropy

Edge entropy anonymisation is proposed by Yan et al. (2018). The algorithm aims to achieve high uncertainty about the edges to obscure the original graph, which they measure via edge entropy. In anonymization, the 10% of the nodes with the highest betweenness centrality are selected. Those with degree larger than 4 are considered to be important nodes. From these important nodes probabilities are assigned to all node pairs in their 1-neighborhoods. From the complete subgraph of the nodes in the neighborhood of an important node, a set of triangles is chosen at random, such that none of their edges overlap. The number of edges in this set, divided by the total possible number of edges in the subgraph, is assigned as the probability for the node pairs in this subgraph. The anonymous graph is constructed by sampling edges using the assigned probabilities.

3.4.13 Link Perturbation

Link perturbation anonymisation is proposed in Mittal et al. (2012) with the goal of hiding node relationships. For each vertex in the graph, a random walk of length 5 from each of its neighbors, and with probability $\frac{0.5*deg(u)-1}{deg(u)-1}$ connect the vertex u with the terminal node of the random walk. If the proposed edge already exists in the graph, a new random walk procedure is done, until either a suitable endpoint is found, or 10 attempts have been made.

3.4.14 Link Mirage

The link mirage algorithm is proposed in Liu and Mittal (2016) with the goal of hiding node relationships. The algorithm works by clustering the graph into several communities, and then perturbing the links within communinities and between communities. The graph is clustered using the maximum modularity method. The edges within communities are perturbed using the same method as in Link Perturbation anonymisation. To perturb the edges between communities, the inter-community edges are selected, and their endpoints changed. For each community c_u, a node u that is the endpoint of a connecting edge is connected to a connecting node v in a different community c_v with probability $\frac{deg(u)*deg(v)*|c_u|}{|E_{uv}|*(|c_u|+|c_v|)}$ where E_{uv} is the set of edges between the two communities.

3.4.15 Random Walk

The random walk anonymisation algorithm is proposed by Guo et al. (2018), with the target of hiding the identity of hub nodes in a network. Nodes that are in the top 10% of degrees are designated as hub nodes. The algorithm does a random walk of length 5 starting at the hub node, and the final node is designated the new hub node. For each hub, neighbors of the original hub are disconnected uniformly at random, and reconnected to the new hub node. This is repeated until the degree of the new hub node is the previous degree of the original hub node. After this shifting of hubs, the overall structure of the graph is perturbed as in Link Perturbation anonymisation.

3.4.16 Upper Approximation based Anonymisation

Upper Approximation based anonymisation is proposed by Kumar and Kumar (2017) with the goal of hiding node relationships. For each node the algorithm collects all nodes in the 2-neighborhood,

and computes their Jaccard similarity based on their direct neighbors. The graph is reconstructed by connecting each node to all the nodes in its 2-neighborhood whose Jaccard similarity is greater than 0.2.

3.4.17 Friend in the Middle

Friend in the Middle (FiM) anonymisation is proposed by Beato et al. (2013) for the purpose of protecting link privacy. To obscure links, a set of nodes from the network is selected uniformly at random, and a subset of the links in the network are redirected through these nodes. Beato et al. (2013) proposed several different methods of selecting the links to obscure. We chose to use one FiM node per link, and to alter a proportion of each node's links. We used 50 FiM nodes, and 20% of each node's edges, since neither obscuring more edges, nor using more FiM nodes produced strongly better privacy results in Beato et al. (2013). The FiM nodes to are selected in a fixed order, so that each FiM node is used equally.

4 RESULTS

In this chapter we present the results of our experiments on the anonymization methods. The details of how we compute the metrics from the anonymous graphs are given in section 3.3.5. We investigate the minimum and average k-anonymity, and the minimum and average differential privacy score to evaluate how fair the privacy is. We investigate the minimum and average number of changed edges, the KS distance to the ground truth degree distribution, the KS distance to the ground truth betweenness centrality distribution, and the KS distance to the ground truth local clustering coefficient distribution to evaluate how fair the utility is. We have chosen to organise this section by algorithm and discuss trends that hold over several algorithms at the beginning.

We found that for k-degree anonymity most of the algorithms show consistent favour towards the smaller communities. This is true both for algorithms that target k-degree anonymity and for those that do not. In the stricter k-anonymity metrics, that is k-count distance 1 and 2, k-VRQ, and k-Isomorphism anonymity, almost all algorithms achieve only 1-anonymity on most of the graphs, and slightly better anonymity on some of the obesity minority groups. The exception is the ENDP anonymisation which achieves 100-anonymity in the soccer graphs. The pattern of performing better only on minority groups in the obesity graphs is likely because these are very small communities with between 11 and 26 nodes, so the chance of them having no unique nodes is significant.

We observe on the differential privacy score, the differential privacy based algorithms tend to achieve better scores than other algorithms. However, a number of the algorithms that do not explicitly target differential privacy achieve similar scores. The differential privacy scores seem to be correlated closely to the denseness of the graphs for many of the anonymisation algorithms, as in fig. 1. We see this behaviour in ANONET degree, ANONET count, ANONET isomorphism, FiM, k-Degree Sequence, KDVEM, Link Mirage, Link Perturbation, OLP, Pseudo Node, Subgraph DP, and Two-Step algorithms, where the denser HBA and Facebook graphs achieve notably better scores than the more sparse obesity and soccer graphs.

We observe the k-anonymity algorithms tend to perform better on the distance to ground truth degree distribution utility than other algorithms. This is likely because many of these algorithms optimise the changes to the graph to create the fewest changes in the degree sequence, which translates directly to the degree distribution. For the algorithms that perform particularly well on the degree distribution utility, that is the FiM, k-Degree Sequence, KDVEM, Link Mirage, Pseudo Node, and Two-Step anonymisation algorithms we also observe that the degree distribution utility is better in larger graphs and majority communities, as seen in fig. 2. This is likely because the number of degree changes required to anonymise a graph or community within a graph increases more slowly than the size of the graph or community, so in larger groups there are relatively fewer degree changes per node, so the overall degree distribution changes less. We observe that the degree distribution utility consistently over algorithms shows more difference between communities on the obesity graphs than on the other graph types, also visible in fig. 2.

We observe that graphs and communities with more low LCC nodes perform better than graphs and communities with fewer low LCC nodes. This is especially visible on the soccer graphs, where most of the nodes have 0 LCC, and where the LCC utility is quite high for most of the algorithms. We observe that the graph structure is a large factor in the betweenness centrality distribution utility. For almost all of the anonymisation algorithms the betweenness centrality distribution utility of the Facebook graph is notably worse than of the other graph types. The algorithms where this is not the case are Edge Entropy, ENDP, KDVEM, and PBCN anonymisation. We do not see a consistent pattern with regards to minority or majority favor in the Facebook graphs. For most of the anonymisation algorithms, that is for the ANNM,

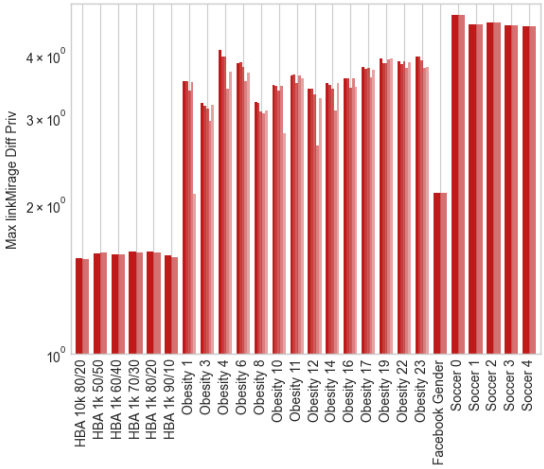

Figure 1. Plot showing the Differential Privacy scores on each of the different graph types anonymised using the Link Mirage algorithm.

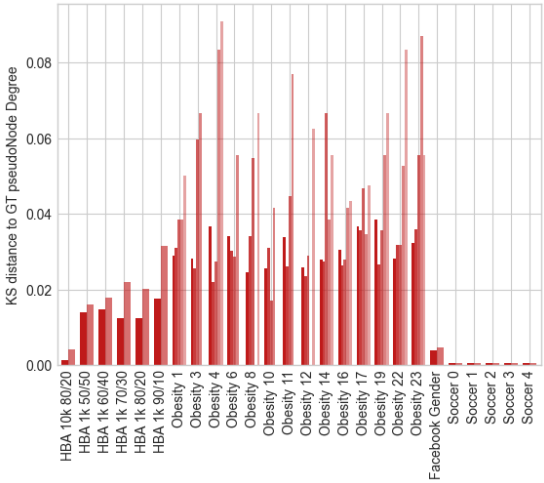

Figure 2. Plot showing the KS distance to the ground truth degree distribution on each of the different graph types anonymised using the Pseudo Node algorithm.

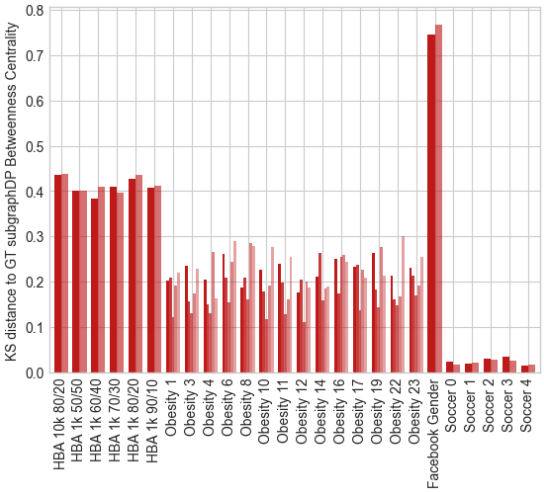

Figure 3. Plot showing the KS distance to the ground truth BC distribution on each of the different graph types anonymised using the subgraphDP algorithm.

ANONET degree, ANONET count, ANONET isomorphism, Edge Entropy, ENDP, Link Mirage, OLP, Random Walk, Upper Set based Anon, and Subgraph DP algorithms we observe that the betweenness centrality distribution utility is notably better on the soccer graphs, as in fig. 3. The top 10 betweenness centrality nodes strongly favors not only minority communities, but also smaller graphs, as seen in fig. 4. This is likely because we use the 10 top nodes regardless of graph or community size, which in smaller groups will represent a larger percentage of the total nodes.

4.1 ANNM

The ANNM algorithm aims to create k-degree anonymity in a graph by adding edges and nodes.

In terms of anonymity, ANNM achieves the given target of 10-degree anonymity for most graphs, and achieves a higher degree anonymity in some minority communities. The minority groups where it performs better are those that have the smallest absolute size, in larger graphs there is no difference between the k-anonymity of communities. This is likely because in communities with very few nodes, it is less unlikely for none of the nodes to be unique, or have low k-anonymity. The differential privacy score more strongly depends on the ground truth graph, with the algorithm performing significantly better in the HBA, Facebook and Soccer graphs than in the obesity graphs.

For the utility, the ANNM algorithm performs similarly in the difference to the Degree distribution, and the LCC distribution. In both metrics, the algorithm performs notably worse on HBA 1k graphs than on the other graphs. While there are differences in the results for different communities, there is no consistent favor towards minority or majority groups. For the difference to the BC distribution, the results are better for larger graphs. On the obesity graphs the smallest communities perform worse than the majority communities, however there is no consistent favoritism towards majority communities on the HBA, Facebook, and Soccer graphs. The mean number of changed edges per node is very low in the obesity and soccer graphs. In the Obesity and Soccer graphs the minority communities have on average fewer changed edges per node, while on the HBA and FB graphs, the reverse is true. The maximum number of changed edges is lower in the Obesity graphs than on the other graphs, and favors the minority communities, where there is no difference in the other graphs. In the number of changed edges per node, there is no consistent difference for different community sizes.

Overall the results that we achieve do not show unfairness with regard to community size in the ANNM algorithm.

4.2 ANONET Degree

In terms of anonymity, ANONET Degree Anonymisation achieves only 1-degree anonymity for most graphs. For the obesity and soccer graphs, ANONET Degree Anonymisation achieves a higher k-anonymity in the minority communities, around 5-degree anonymity. In the other k-anonymity metrics

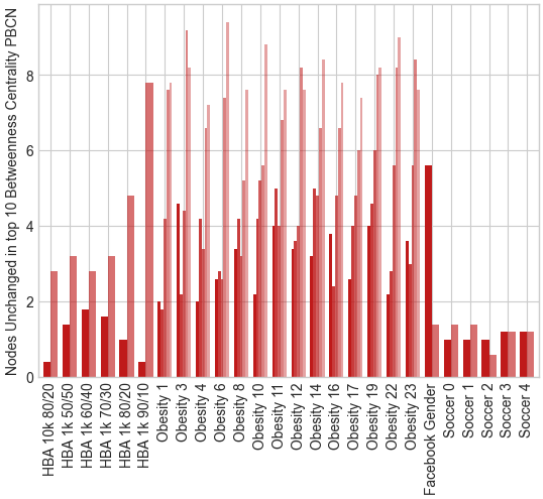

Figure 4. Plot showing how many of the top 10 BC nodes remain unchanged on each of the different graph types anonymised using the PBCN algorithm.

it achieves only 1-anonymity, and 2-anonymity in some of the minority communities. The differential privacy score from ANONET Degree Anonymisation depends strongly on the graph structure. The HBA and Facebook graphs achieve a DP score of around 2, while the obesity and soccer graphs have a worse DP score of around 5. There is no difference between the differential privacy score based on community size. In the degree distribution utility, ANONET Degree Anonymisation performs consistently for different graph types, except on the soccer graphs. One the soccer graphs the degree distribution utility of the majority communities is worse than on other graph types, while the utility on the minority communities is better. While there are differences in the degree distribution utility for different communities on the HBA, obesity, and Facebook graphs, there is no consistent favour towards either minority or majority communities. In the local clustering utility, ANONET Degree Anonymisation performs consistently for different graph types, except on the soccer graphs, where the utility is much better. While there are differences in the local clustering utility for different communities, there is no consistent favour towards either minority or majority communities. In the betweenness centrality distribution change, ANONET Degree Anonymisation performs consistently for different graph types, except on the Facebook graph, where the utility is much worse. While there are differences in the betweenness centrality distribution change for different communities, there is no consistent favour towards either minority or majority communities. Considering the top 10 betweenness centrality nodes, ANONET Degree Anonymisation performs best in very small communities. In the soccer graphs none of the top 10 BC nodes are maintained.

The mean number of changed edges per node is very low in the obesity and soccer graphs. In these graphs the minority communities have on average slightly fewer changed edges per node, while on the HBA and FB graphs, the reverse is true. The maximum number of changed edges per node favors the minority communities.

Overall the results that we achieve do not show unfairness with regard to community size in the ANONET Degree Anonymisation algorithm.

4.3 ANONET Count

In terms of anonymity, ANONET Count Anonymisation achieves only 1-degree anonymity for most graphs. For the obesity and soccer graphs, ANONET Count Anonymisation achieves a higher k-anonymity in the minority communities, around 5-degree anonymity. In the other k-anonymity metrics it achieves only 1-anonymity, and 2-anonymity in some of the minority communities. The differential privacy score from ANONET Count Anonymisation depends strongly on the graph structure. The HBA and Facebook graphs achieve a DP score of around 2, while the obesity and soccer graphs have a worse DP score of around 4.5. There is no difference between the differential privacy score based on community size. In the degree distribution utility, ANONET Count Anonymisation performs consistently for different graph types, except on the soccer graphs. One the soccer graphs the degree distribution utility of the

majority communities is worse than on other graph types, while the utility on the minority communities is better. While there are differences in the degree distribution utility for different communities on the HBA, obesity, and Facebook graphs, there is no consistent favour towards either minority or majority communities. In the local clustering utility, ANONET Count Anonymisation performs consistently for different graph types, except on the soccer graphs, where the utility is much better. While there are differences in the local clustering utility for different communities, there is no consistent favour towards either minority or majority communities. In the betweenness centrality distribution change, ANONET Count Anonymisation performs consistently for different graph types, except on the Facebook graph, where the utility is much worse. While there are differences in the betweenness centrality distribution change for different communities, there is no consistent favour towards either minority or majority communities. Considering the top 10 betweenness centrality nodes, ANONET Count Anonymisation performs best in very small communities. In the soccer graphs none of the top 10 BC nodes are maintained.

The mean number of changed edges per node is very low in the obesity and soccer graphs. In these graphs the minority communities have on average slightly fewer changed edges per node, while on the HBA and FB graphs, the reverse is true. The maximum number of changed edges per node favors the minority communities.

Overall the results that we achieve do not show unfairness with regard to community size in the ANONET Count Anonymisation algorithm.

4.4 ANONET dk-Isomorphism

In terms of anonymity, ANONET dk-Isomorphism Anonymisation achieves only 1-degree anonymity for most graphs. For the obesity and soccer graphs, ANONET dk-Isomorphism Anonymisation achieves a higher k-anonymity in the minority communities, around 5-degree anonymity. In the other k-anonymity metrics it achieves only 1-anonymity, and 2-anonymity in some of the minority communities. The differential privacy score from ANONET dk-Isomorphism Anonymisation depends strongly on the graph structure. The HBA and Facebook graphs achieve a DP score of around 2, while the obesity and soccer graphs have a worse DP score of around 4. There is no difference between the differential privacy score based on community size. In the degree distribution utility, ANONET dk-Isomorphism Anonymisation performs consistently for different graph types, except on the soccer graphs. One the soccer graphs the degree distribution utility of the majority communities is worse than on other graph types, while the utility on the minority communities is better. While there are differences in the degree distribution utility for different communities on the HBA, obesity, and Facebook graphs, there is no consistent favour towards either minority or majority communities. In the local clustering utility, ANONET dk-Isomorphism Anonymisation performs consistently for different graph types, except on the soccer graphs, where the utility is much better. While there are differences in the local clustering utility for different communities, there is no consistent favour towards either minority or majority communities. In the betweenness centrality distribution change, ANONET dk-Isomorphism Anonymisation performs consistently for different graph types, except on the Facebook graph, where the utility is much worse. While there are differences in the betweenness centrality distribution change for different communities, there is no consistent favour towards either minority or majority communities. Considering the top 10 betweenness centrality nodes, ANONET dk-Isomorphism Anonymisation performs best in very small communities. In the soccer graphs none of the top 10 BC nodes are maintained.

The mean number of changed edges per node is very low in the obesity and soccer graphs. In these graphs the minority communities have on average slightly fewer changed edges per node, while on the HBA and FB graphs, the reverse is true. The maximum number of changed edges per node favors the minority communities.

Overall the results that we achieve do not show unfairness with regard to community size in the ANONET dk-Isomorphism Anonymisation algorithm.

4.5 DPFT

In terms of anonymity, DPFT achieves only 1 for k-anonymity. For some of the minority groups, DPFT achieves a slightly higher k-anonymity, around 3. For the differential privacy score, DPFT achieves a score of just less than 4 on the obesity graphs, and around 2 on the other graph types. DPFT performs worse on the obesity graphs than on the other graph types. There is for most graph types no difference between the differential privacy score based on community size. On the obesity graphs the minority groups tend to perform slightly better.

For the utility DPFT performs well on the degree distribution utility. In the degree distribution utility, DPFT performs better on the obesity and Facebook graphs than on the HBA and soccer graphs. The majority communities tend to maintain better degree distribution utility than the minority communities.

DPFT shows similar patterns in the betweenness centrality, and local clustering utilities. The utility is notably worse for the Facebook graph than for other graph types. While there are differences in the results of the local clustering utility for different communities, there is no consistent favour towards either minority or majority communities. In the ranked betweenness centrality there is favor towards minority communities, likely because we use the top 10 betweenness nodes, which is a larger proportion in groups with fewer nodes. The mean number of changed edges per node is very low in the obesity and soccer graphs. In these graphs the minority communities have on average slightly fewer changed edges per node, while on the HBA and FB graphs, the reverse is true. The maximum number of changed edges favors the minority communities.

Overall the results that we achieve show worse results for minority groups only in the degree distribution utility, but otherwise do not show unfairness with regard to community size in the DPFT algorithm. This indicates that there might be some applications where DPFT anonymisation will cause unfair results, but not all applications.

4.6 Edge Entropy

In terms of anonymity, Edge Entropy achieves only 1 for k-anonymity. In the minority communities of the obesity graphs, it achieves a notably higher k-anonymity, around 30. For the differential privacy score, Edge Entropy achieves a score of around 2, and performs worse on the obesity graphs than on the other graph types. There is for most graph types no difference between the differential privacy score based on community size. On the obesity graphs the minority groups tend to perform slightly better.

For the utility Edge Entropy performs very well on the soccer graphs. In the degree distribution utility, Edge Entropy performs consistently for different graph types, except for the soccer graphs. While there are differences in the degree distribution utility for different communities, there is no consistent favour towards either minority or majority communities. For the local clustering utility, Edge Entropy performs better on the HBA and Facebook graphs than on the Obesity graphs, and again very well on the soccer graphs. While there are differences in the local clustering utility for different communities, there is no consistent favour towards either minority or majority communities. In the betweenness centrality distribution change, Edge Entropy performs fairly consistently for different graph types, except on the Soccer graphs where it does significantly better. While there are differences in the betweenness centrality distribution change for different communities, there is no consistent favour towards either minority or majority communities. Considering the top 10 betweenness centrality nodes, Edge Entropy performs best in very small communities, although the difference is less pronounced than in other algorithms. In the soccer graphs none of the top 10 BC nodes are the same.

The mean number of changed edges per node is very low in the obesity and soccer graphs. In these graphs the minority communities have on average slightly fewer changed edges per node, while on the HBA and FB graphs, the reverse is true. The maximum number of changed edges per node favors the minority communities.

Overall the results that we achieve do not show unfairness with regard to community size in the Edge Entropy algorithm.

4.7 ENDP

For the ENDP algorithm the k-anonymity is strongly dependent on the graph structure. The ENDP anonymised Facebook graph achieved only 1-degree anonymity, while the soccer graphs have a k-degree anonymity of over 10000. For the differential privacy score ENDP achieves a score of around 1.6, which is consistent over different graph types. The minority communities consistently have a slightly better differential privacy score.

For the utility the results for ENDP anonymisation differ strongly by graph type. In the degree distribution utility, ENDP performs better on the 1k HBA and obesity graphs, than on the other graph types. In the HBA and soccer graphs, the majority communities tend to perform better, while in the obesity and Facebook graphs the minority communities maintain better degree distribution utility. For both the degree distribution utility and the local clustering utility, ENDP has very poor results on the Facebook graph. In the local clustering utility, ENDP performs well on the obesity and soccer networks, and poorly

on the HBA and Facebook graphs. While there are differences in the results of the local clustering utility for different communities, there is no consistent favour towards either minority or majority communities.

In the betweenness centrality distributuion difference, ENDP seems to perform better for the larger graphs. In the Facebook and Soccer graphs the minority communities perform better, while in the HBA and obesity graphs the majority communities maintain a betweenness centrality distribution more similar to the original graphs. Considering the top 10 betweenness centrality nodes, ENDP performs best in very small communities. In the 10k HBA, Facebook, and soccer graphs, none of the top 10 BC nodes are the same.

The mean number of changed edges per node is very low in the obesity and soccer graphs. In these graphs the minority communities have on average slightly fewer changed edges per node, while on the HBA and FB graphs, the reverse is true. The number of changed edges favors the minority communities.]

Overall the results that we achieve are inconsistent over graph types, but seem to show favour towards minority communities when using ENDP anonymisation.

4.8 FiM

In terms of anonymity, FiM achieves only 1 k-anonymity. For some of the minority groups, FiM achieves a slightly higher k-degree anonymity, around 5. For the differential privacy score, FiM performs quite well, and achieves a DP score of 1.4 on the HBA graphs, and around 1.8 on the other graph types. There is very little difference between the differential privacy score based on community size, but in the HBA and obesity graphs the minority communities have a slightly better differential privacy score.

For the utility FiM performs well on the degree distribution utility, as the algorithm changes the degree of only a few nodes. FiM also performs generally well on the soccer graphs. In the degree dsitribution utility, FiM performs better on the larger 10k HBA, soccer and Facebook graphs than on the smaller graphs. While there are differences in the degree distribution utility for different communities, there is no consistent favour towards either minority or majority communities. In the local clustering utility, FiM performs consistetnly over the different graph types, except for the soccer graphs, where it does significantly better. The minority communities tend to maintain better local clustering utility than the majority communities.

In the betweenness centrality distributuion change, FiM performs consistetnly over the different graph types, except for the Facebook graph, where it does significantly worse. While there are differences in the betweeness centrality distribution change for different communities, there is no consistent favour towards either minority or majority communities. Considering the top 10 betweenness centrality nodes, FiM performs best in very small communities. Except for in the obesity graphs, between 0 and 2 of the top 10 BC nodes are the same.

The mean number of changed edges per node is very low in the obesity and soccer graphs. In these graphs the minority communities have on average slightly fewer changed edges per node, while on the HBA and FB graphs, the reverse is true. There is no consistent difference across community sizes in the maximum number of changed edges per node.

Overall the results that we achieve indicate that the FiM algorithm creates slightly better privacy in minority communities. We do not obeserve a difference between minority and majority results in the utility metrics, although the results to vary strongly by graph structure.

4.9 k-Degree Sequence Anonymisation

In terms of anonymity, k-Degree Sequence Anonymisation achieves the given target of 10-degree anonymity for most graphs, and a higher degree anonymity around 20 in the obesity minority communities. For the differential privacy score, k-Degree Sequence Anonymisation has a DP score of around 4 for the obesity and soccer graphs. However, k-Degree Sequence Anonymisation gives a very good DP score of close to 1 for the HBA graphs. There is no difference in the differential privacy score based on community size.

For the utility k-Degree Sequence Anonymisation performs well on all of the utility metrics. k-Degree Sequence Anonymisation performs similarly in the degree distribution and local clustering utility. The larger soccer and HBA 10k graphs have better utility than the smaller obesity and HBA 1k graphs. The majority communities consistently to maintain better degree distribution and local clustering utility than the minority communities. In the betweenness centrality distributuion change, k-Degree Sequence Anonymisation performs better on the HBA and soccer graphs than on the obesity and Facebook graphs. In the HBA, obesity, and Facebook graphs the majority communities have better betweeness centrality

distribution utility, slightly in the HBA and Facebook graphs and significantly in the obesity graphs. There is no consistent difference for different community sizes in the soccer graphs. Considering the top 10 betweenness centrality nodes, *k*-Degree Sequence Anonymisation performs quite consistently over graph types. It consistently performs better in minority communities. The mean number of changed edges per node is very low in the obesity, Facebook, and soccer graphs. In these graphs the minority communities have on average slightly fewer changed edges per node. On the HBA graph there is no consistent difference for community sizes. The maximum number of changed edges for a node favors the minority communities.

Overall the results that we achieve indicate that the *k*-Degree Sequence Anonymisation algorithm creates slightly better utility in minority communities. We do not obeserve a difference between minority and majority results in the privacy metrics, although very small communities are favored in k-degree anonymity.

4.10 KDVEM

In terms of anonymity, KDVEM achieves the specified target of 10-degree anonymity. For some of the minority groups, KDVEM achieves a slightly higher k-anonymity, around 20. For the differential privacy score, KDVEM performs notably better on the HBA and Facebook graphs than on the soccer and obesity graphs. The DP score for the HBA graphs is around 1.5, and that for the Facebook graph about 2, while the DP score for the other graphs is around 3. There is no difference between the differential privacy score based on community size.

For the utility KDVEM performs well on all of the utility metrics. In the degree distribution utility, KDVEM performs better on the larger HBA 10k, Facebook, and soccer graphs, than on the HBA 1k and obesity graphs. While there are differences in the degree distribution utility for different communities, there is no consistent favour towards either minority or majority communities.

In the local clustering utility, KDVEM performs similarly as in the degree distribution utility; better on the larger HBA 10k, Facebook, and soccer graphs, than on the HBA 1k and obesity graphs. In the HBA, and Facebook graphs the majority communities perform notably better, while in the soccer graphs the minority communities maintain better local clustering utility. There is no consistent difference for different community sizes in the obesity graphs.

In the betweenness centrality distributuion change, KDVEM performs better on the HBA and soccer graphs and worse on the obesity and Facebook graphs. In the obesity and Facebook graphs, the majority communities tend to perform better, while in the HBA and soccer graphs there is no difference for different community sizes. Considering the top 10 betweenness centrality nodes, KDVEM performs best in very small communities. In the HBA and soccer graphs only between 1 and 4 of the top 10 BC nodes are the same.

The mean number of changed edges per node is very low in the obesity, Facebook, and soccer graphs. In these graphs the minority communities have on average slightly fewer changed edges per node, while on the HBA graphs, there is no consistent pattern. In the maximum number of changed edges per node, there is no consistent difference for different community sizes.

Overall the results that we achieve do not show unfairness with regard to relative community size in privacy or utility, although we do observe that the k-anonymity is better in communities and graphs with very few nodes.

4.11 Link Mirage

In terms of anonymity, Link Mirage achieves only 1 for k-degree anonymity. For some of the minority groups, Link Mirage achieves a slightly higher k-anonymity, around 15. For the differential privacy score, Link Mirage achieves a value of around 2 on the HBA and Facebook graphs, and a value of around 4 on the soccer and obesity graphs. There is no difference between the differential privacy score based on community size.

For the utility Link Mirage performs better on the soccer graphs than on the other graph types. In the degree distribution utility, Link Mirage performs fairly consistently over different graph types, with slightly better results on the soccer graphs. The majority communities tend to maintain better degree distribution utility than the minority communities. In the local clustering utility, Link Mirage performs consistently for the HBA and obesity graphs, but worse on the Facebook graph, and better on the soccer graphs. While there are differences in the local clustering utility for different communities, there is no consistent favour towards either minority or majority communities. In the betweenness centrality

distributuion change, Link Mirage performs consistently for the HBA and obesity graphs, but worse on the Facebook graph, and better on the soccer graphs. While there are differences in the betweenness centrality distributuion change for different communities, there is no consistent favour towards either minority or majority communities. Considering the top 10 betweenness centrality nodes, Link Mirage performs best in very small communities, and better in minority communities than in majority communities. The mean number of changed edges per node is very low in the obesity and soccer graphs. In these graphs there is no difference between the results for different community sizes, while on the HBA and FB graphs, the majority communities perform better. The maximum number of changed edges per node favors the minority communities.

Overall the results that we achieve do not show unfairness with regard to community size in the Link Mirage algorithm, although the results to vary strongly by graph structure.

4.12 Link Perturbation

In terms of anonymity, Link Perturbation achieves only 1 degree anonymity. For some of the minority groups, Link Perturbation achieves a slightly higher k-anonymity, around 10. For the differential privacy score, Link Perturbation achieves a value of around 3 on the obesity, Facebook, and soccer graphs, and about 1.5 on the HBA graphs. There is no difference between the differential privacy score based on community size.

For the utility Link Perturbation performs well on the degree distribution utility. In the degree distribution utility, Link Perturbation performs better on the obesity and Facebook graphs, and worse on the HBA and soccer graphs. In the HBA and soccer graphs the minority communities tend to perform better, while in the obesity and Facebook graphs the majority communities maintain better degree distribution utility. In the local clustering utility, Link Perturbation performs consistently for the HBA and obesity graphs, but worse on the Facebook graph, and better on the soccer graphs. While there are differences in the local clustering utility for different communities, there is no consistent favour towards either minority or majority communities.

In the betweenness centrality distributuion change, Link Perturbation performs consistently for the HBA, obesity, and soccer graphs, but performs worse on the Facebook graph. There is no difference in the betweenness centrality distributuion change between minority and majority communities. Considering the top 10 betweenness centrality nodes, Link Perturbation performs best in very small communities.

The mean number of changed edges per node is very low in the obesity and soccer graphs. In these graphs the minority communities have on average slightly fewer changed edges per node, while on the HBA and FB graphs, the reverse is true. The maximum number of changed edges per node favors the minority communities.

Overall the results that we achieve do not show unfairness with regard to community size in the Link Perturbation algorithm. We do obeserve differences between minority and majority results in [the utility metrics, although whether minority or majority communities perform better varies strongly by graph structure.

4.13 OLP

In terms of anonymity, OLP achieves only 1 for k-degree anonymity. For some of the minority groups, OLP achieves a slightly higher k-anonymity, around 10. For the differential privacy score, OLP achieves a value of around 2 on the HBA and Facebook graphs, and a value of around 6 on the soccer and obesity graphs. There is no difference between the differential privacy score based on community size.

In the degree distribution utility, OLP performs quite consistently for different graph types, with slightly worse results on the HBA graphs than on the other graph types. On the Facebook and soccer graphs, the minority communities maintain better degree distribution utility than the majority communities. On the HBA and obesity graphs there is no consistent difference in degree distribution utility between community sizes. In the local clustering utility, OLP performs better on the soccer graphs than on the other graph types. While there are differences in the local clustering utility for different communities, there is no consistent favour towards either minority or majority communities. In the betweenness centrality distributuion change, OLP performs consistently for different graph types, except the Facebook graph where it performs notably worse. In the obesity and Facebook graphs themajority communities tend to perform better, while in the HBA and soccer graphs there is no difference in betweenness centrality distributuion utility between community sizes. Considering the top 10 betweenness centrality nodes, OLP performs best in very small communities. In the soccer graphs, the majority communities do not

maintain any of the top 10 BC nodes. The mean number of changed edges per node is very low in the obesity and soccer graphs. In the soccer and Facebook graphs the minority communities have on average slightly fewer changed edges per node, while on the HBA and obesity graphs, there is no consistent difference between community sizes. The maximum number of changed edges per node favors the minority communities.

Overall the results that we achieve indicate that the OLP algorithm creates slightly better utility in minority communities, however this depends strongly on the graph structure.

4.14 PBCN

In terms of anonymity, PBCN achieves only 1-degree anonumity. For some of the minority groups, PBCN achieves a slightly higher k-anonymity, around 5. For the differential privacy score, PBCN performs quite well, achieving a DP score of around 1.1 on the HBA and soccer graphs, and around 1.6 on the obesity and Facebook graphs. The minority communities consistently have a slightly better differential privacy score.

In the degree distribution utility, PBCN performs worse on the soccer graphs than on the other graph types. In the obesity and soccer graphs the majority communities tend to perform better, while in the HBA and Facebook graphs there is no consistent favor towards either minority or majority communities. In the local clustering utility, PBCN performs fairly consistently for different graph types, although the results are slightly worse on the Facebook graph and slightly better on the soccer graphs. While there are differences in the local clustering utility for different communities, there is no consistent favour towards either minority or majority communities. In the betweenness centrality distributuion change, PBCN performs worse on the soccer and Facebook graphs than on the HBA and obesity graphs. The majority communities tend to maintain better betweenness centrality distributuion utility than the minority communities. Considering the top 10 betweenness centrality nodes, PBCN performs best in very small communities. In the HBA and soccer graphs very few of the top 10 BC nodes are maintained.

The mean number of changed edges per node is very low in the obesity and soccer graphs. In these graphs there is no favor towards either minority or majority communities. On the HBA and FB graphs, the majority graphs have on average slightly fewer changed edges per node. The maximum number of changed edges per node favors the minority communities.

Overall the results that we achieve indicate that the PBCN algorithm creates slightly better privacy in minority communities, and slightly better utility in majority communities.

4.15 Pseudo Node Anonymisation

In terms of anonymity, Pseudo Node Anonymisation achieves the given target of 10-degree anonymity. For some of the minority groups, Pseudo Node Anonymisation achieves a slightly higher k-degree anonymity, around 20. For the other k-anonymity metrics, Pseudo Node Anonymisation achieves only 1-anonymity, with slightly higher values in some of the very small communities. For the differential privacy score, Pseudo Node Anonymisation performs notably better on the HBA graphs, with a DP score of around 1.5, while on the other graph types it achieves a DP score around 3. There is no difference between the differential privacy score based on community size.

For the utility Pseudo Node Anonymisation performs very well on the degree distribution utility.

The degree distribution utility of the graphs anonymised with Pseudo Node Anonymisation seems to be directly correlated to the size of the graph. Pseudo Node Anonymisation performs the best on the very large soccer graphs, and worst on the obesity graphs. In line with this the majority communities tend to maintain better the degree distribution utility, than the minority communities. In the local clustering utility, Pseudo Node Anonymisation performs better on the HBA10k, obesity, Facebook, and soccer graphs, and worse on the smaller HBA 1k graphs. While there are differences in the local clustering utility for different communities, there is no consistent favour towards either minority or majority communities. In the betweenness centrality distributuion change, Pseudo Node Anonymisation performs quite consistently for different graph types. In the obesity and Facebook graphs the majority communities tend to perform better, while in the HBA and soccer graphs there is no favor towards either minority or majority communities. Considering the top 10 betweenness centrality nodes, Pseudo Node Anonymisation performs quite well. It performs best in very small communities, but even the larger soccer graphs consistently maintain around 2 of the same top 10 BC nodes.

The mean number of changed edges per node is very low in the obesity and soccer graphs. In these graphs the minority communities have on average slightly fewer changed edges per node. On the Facebook

graph the minority community has a much lower average changed edges per node. On the HBA graphs, there is no consistent difference between community sizes. In the maximum number of changed edges per node, there is no consistent difference for different community sizes.

Overall the results that we achieve do not show unfairness with regard to community size in the Pseudo Node Anonymisation algorithm.

4.16 Random Walk Anonymisation

In terms of anonymity, Random Walk Anonymisation achieves only 1-degree anonymity. For some of the minority groups, Random Walk Anonymisation achieves a slightly higher k-anonymity, around 4. For the differential privacy score, Random Walk Anonymisation performs best on the HBA graphs with a value of around 1.2, and worst on the soccer graphs with a value of about 2, where the obesity and Facebook graphs achieve a DP score of about 1.6. There is no difference between the differential privacy score based on community size.

For the utility Random Walk Anonymisation performs better on the soccer graphs than on the other graph types. In the degree distribution utility, Random Walk Anonymisation performs consistently for different graph types, and there is no difference in the degree distribution utility between minority and majority communities. In the local clustering utility, Random Walk Anonymisation performs fairly consistently for different graph types, although worse on the Facebook graph and better on the Soccer graphs. The minority communities tend to maintain better local clustering utility than the majority communities. In the betweenness centrality distribution change, Random Walk Anonymisation performs fairly consistently for different graph types, although worse on the Facebook graph and better on the Soccer graphs. While there are differences in the betweenness centrality distribution change for different communities, there is no consistent favour towards either minority or majority communities. Considering the top 10 betweenness centrality nodes, Random Walk Anonymisation performs best in very small communities. In the HBA, Facebook, and Soccer graphs on average less than 1 of the top 10 BC nodes are the same.

The mean number of changed edges per node is very low in the obesity and soccer graphs. In these graphs there is no difference between the minority and majority communities, while on the HBA and FB graphs, majority communities have slightly fewer average changed edges per node. The maximum number of changed edges per node favors the minority communities.

Overall the results that we achieve indicate that the Random Walk Anonymisation algorithm creates slightly better utility in minority communities. We do not obeserve a difference in results based on relative group size in the privacy metrics, although we do see better k-anonymity in communities with very small absolute size.

4.17 Upper Approximation based Anonymisation

In terms of anonymity, Upper Approximation based Anonymisation achieves only 1-degree anonymity. For some of the minority groups, Upper Approximation based Anonymisation achieves a slightly higher k-anonymity, around 10. For the differential privacy score, Upper Approximation based Anonymisation performs worst on the obesity graphs, with a DP score of about 3, quite well on the HBA and Facebook graphs with a DP score around 1.5, and very well on the soccer graphs, where the DP score is nearly 1. The minority communities consistently have a slightly better differential privacy score.

For the utility Upper Approximation based Anonymisation performs poorly on the degree distribution and local clustering utilities. In the degree distribution utility, Upper Approximation based Anonymisation performs better on the obesity and Facebook graphs and worse on the HBA and soccer graphs. While there are differences in the degree distribution utility, for different communities, there is no consistent favour towards either minority or majority communities. In the local clustering utility, Upper Approximation based Anonymisation performs consistently for different graph types. While there are differences in the local clustering utility for different communities, there is no consistent favour towards either minority or majority communities. In the betweenness centrality distribution change, Upper Approximation based Anonymisation performs better on the obesity and soccer graphs and worse on the HBA and Facebook graphs. In the obesity graphs the majority communities tend to perform better, while on the other graph types there is no consistent favor towards either minority or majority communities. Considering the top 10 betweenness centrality nodes, Upper Approximation based Anonymisation performs quite poorly overall. In the HBA graphs, and in the majority communities for most of the obesity and soccer graphs none of the top 10 BC nodes are the same.

The mean number of changed edges per node is very low in the obesity graphs. On the soccer graphs the minority communities have on average slightly fewer changed edges per node, while on the other graph types there is no consistent favor towards minority or majority communities. The maximum number of changed edges per node slightly favors the minority communities.

Overall the results that we achieve do not show unfairness with regard to community size in the Upper Approximation based Anonymisation algorithm. We do obeserve differences between minority and majority results in a number of the utility metrics, although whether minority or majority communities perform better varies strongly by graph structure.

4.18 Subgraph DP

In terms of anonymity, Subgraph DP achieves only 1-degree anonymity. For some of the minority groups, Subgraph DP achieves a slightly higher k-anonymity, around 10. For the differential privacy score, Subgraph DP performs quite consistently over graph types, and achieves a score of about 3, except on the HBA graphs where the DP score is about 1.5. There is no difference between the differential privacy score based on community size.

For the utility Subgraph DP performs poorly on the Facebook graph. In the degree distribution utility, Subgraph DP performs better on the obesity and Facebook graphs, slightly worse on the soccer graphs, and worst on the HBA graphs. In the obesity, Facebook, and soccer graphs the majority communities tend to perform better, while in the HBA graphs the minority communities maintain better the degree distribution utility. In the local clustering utility, Subgraph DP performs fairly consistently for different graph types, with slightly worse results on the Facebook graph and better results on the soccer graphs. In the soccer and obesity graphs the minority communities tend to perform slightly better, while in the HBA and Facebook graphs there is no difference between the majority and minority community results. In the betweenness centrality distribution change, Subgraph DP performs similarly as in the local clustering utility, the Facebook graph has the worst utility, and the soccer graphs have the best. While there are differences in the betweenness centrality distribution change for different communities, there is no consistent favour towards either minority or majority communities. Considering the top 10 betweenness centrality nodes, Subgraph DP performs best in very small communities.

The mean number of changed edges per node is very low in the obesity and soccer graphs. In these graphs the minority communities have on average slightly fewer changed edges per node, while on the HBA and FB graphs, the reverse is true. The maximum number of changed edges per node favors the minority communities.

Overall the results that we achieve do not show unfairness with regard to community size. We do obeserve some differences between minority and majority results in, although whether minority or majority communities perform better varies strongly by graph structure.

4.19 twoStep

In terms of anonymity, Two Step Anonymisation achieves the target of 10-degree anonymity. For some of the minority groups, Two Step Anonymisation achieves a slightly higher k-anonymity, around 20. In the other k-anonymity metrics, Two Step Anonymisation achieves only 1-anonymity, with some of the minority communities achieving around 4-anonymity. For the differential privacy score, Two Step Anonymisation achieves a DP score of about 3 on the obesity, Facebook, and soccer graphs, and a score of about 1.5 on the HBA graphs. The minority communities consistently have a slightly better differential privacy score.

In the degree distribution utility, Two Step Anonymisation performs better on the larger soccer, Facebook, and HBA 10k graphs than on the smaller obesity and HBA 1k graphs. The majority communities tend to maintain better the degree distribution utility than the minority communities. In the local clustering utility, Two Step Anonymisation performs consistently for different graph types. While there are differences in the local clustering utility for different communities, there is no consistent favour towards either minority or majority communities on the HBA and obesity graphs. On the Facebook graph Two Step Anonymisation slightly favors the majority community, while on the soccer graphs it favors the minority communities. In the betweenness centrality distribution change, Two Step Anonymisation performs better on the HBA and soccer graphs and worse on the obesity and Facebook graphs. In the HBA, obesity, and Facebook graphs the majority communities tend to perform better, while on the soccer graphs the minority communities maintain better betweenness centrality distribution utility. Considering the top 10 betweenness centrality nodes, Two Step Anonymisation performs best in very small communities.

The mean number of changed edges per node is very low in the obesity and soccer graphs. In the obesity, Facebook, and soccer graphs the minority communities have on average fewer changed edges per node, while on the HBA graphs, the reverse is true. In the maximum number of changed edges per node there is no consistent difference for different community sizes.

Overall the results that we achieve do not show unfairness with regard to community size in the Two Step Anonymisation algorithm. We do obeserve differences between minority and majority results in some metrics although whether minority or majority communities perform better varies strongly by graph structure and metric.

5 CONCLUSION

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